Unit-IV: Semiconductors

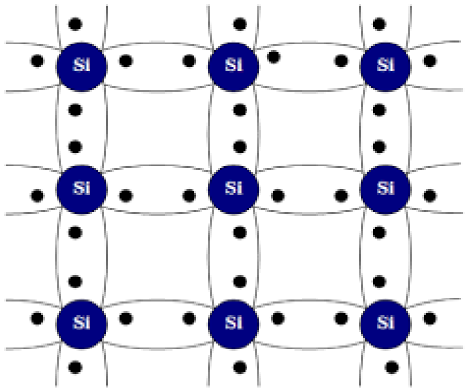
Semiconductors

Semiconductors are materials whose conductivity is in the range between that of good conductors and insulators. **In semiconductors the valence band and conduction band are separated by a small energy gap** (less than 5eV). Hence at low temperature semiconductors behaves as insulators and as temperature increases, the conductivity increases. The resistivity of semiconductors lies in the range 10^{-6} to 10^{+8} Ω m. These intermediate properties are determined by the crystal structure, bonding characteristics, and electronic energy bands etc. elements such as silicon (Si, Eg=1.1eV), germanium (Ge, Eg=0.72eV), and compounds such as gallium arsenide (GaAs, Eg=1.4eV), gallium phosphide (GaP, Eg=2.26eV) are examples of semiconductors.

Classification of Semiconductors:

Semiconductors are classified as intrinsic and extrinsic semiconductors.

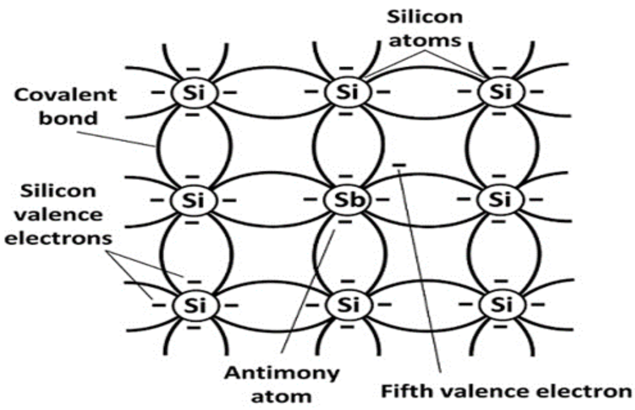
Intrinsic (Pure) semiconductors: A **semiconductor in its pure form, which does not have any external impurity is known as intrinsic semiconductor.** In intrinsic semiconductor, even at room temperature, electron – hole pairs are created. To understand this, let us consider the arrangement of atoms in semiconducting material such as silicon. The carriers contributing to electrical conduction are generated due to the breaking of covalent bonds by thermal activation process. Pure germanium or silicon are examples of intrinsic semiconductors. The silicon having 4 electrons in outermost shell that constitute the valence electron. These materials have four valence electrons and require four more to complete the subshell. Hence, they tend to form four covalent bonds with neighbouring atoms. This is shown schematically in two dimensions in Fig.

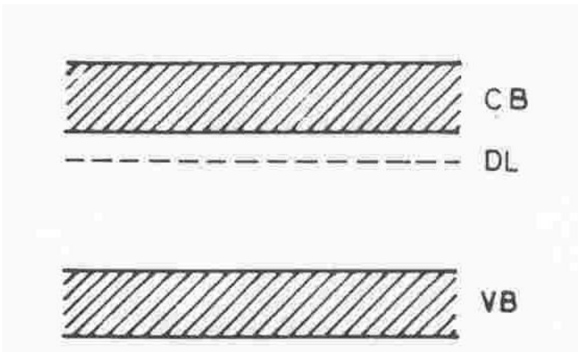


Extrinsic semiconductor: **A semiconductor to which an impurity is added or doped with some impurities are known as extrinsic semiconductor or doped semiconductor**. Conductivity of these semiconductors increases due to increasing the doping concentration of electron or holes. The doped semiconductor or extrinsic semiconductor is either n-type or p-type depending on the valence of doping agents. The added impurities are called dopants. The impurities normally employed for germanium and silicon are elements of group III or group V of the periodic table. Boron, gallium and indium from group III and phosphorus, arsenic and antimony from group V are the common dopants. The extrinsic are classified as **n-type or donor type** and ***p*-type or acceptor typ**e semiconductors.

n-type semiconductors:

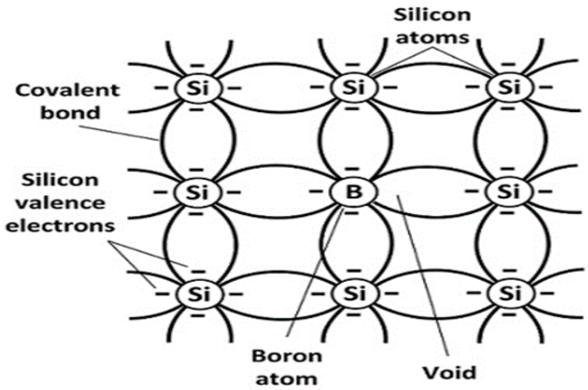
When a small amount of pentavalent impurity is added to a pure semiconductor, it forms n-type semiconductor. the addition of impurity increases electron concentration, so these are known as **donor impurities**, since they provide free electrons to the semiconductor crystal. Consider a pure silicon crystal, which has four valence electrons. When a pentavalent impurity is added such as arsenic or antimony with five valence electrons, four valence electrons of antimony fit in the silicon crystal to form covalent bonds, but one electron of impurity atom is left free. Hence each impurity atom added contributes one free electron to the crystal, thereby creating surplus electron in the Si crystal. This free electron may travel from valence band to conduction band. Since the concentration of charge carrying electrons is more than that of holes, these semiconductors are called n-type, where "n" stands for negative. When impurity atoms are added into the lattice of semiconductors, new energy levels are introduced in the band structure. The energy state of this electron is represented by an energy level just below the bottom of the conduction band called donor level.

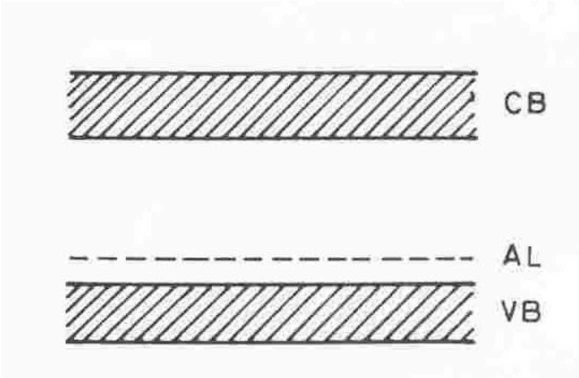




p-type semiconductors:

When a small amount of trivalent impurity is added to a pure semiconductor, it forms a p-type semiconductor. The addition of trivalent impurity creates more holes. Since these holes created can accept any electron, these impurities are called **accepter impurities**. Consider pure silicon crystal to which a trivalent impurity such as Boron or Gallium (with 3 valence electrons) is added, and each of which combines with three valence electrons of three silicon atoms to form covalent bonds. But the fourth bond is incomplete, and a vacancy is left free, and this forms a hole. Hence each impurity atom added there creates a hole as shown in fig. Due to this, there exist a large number of holes in the silicon crystal. Here some electrons move to conduction band due to thermal breakage of covalent bonds, but the holes are more than these electrons. Hence here current conduction is due to the holes which are positively charged. **Acceptor levels** lie close to the top of the valence band and capture electrons from the covalent bonds thereby generating holes in the valence band. The energy band structure as shown in fig.



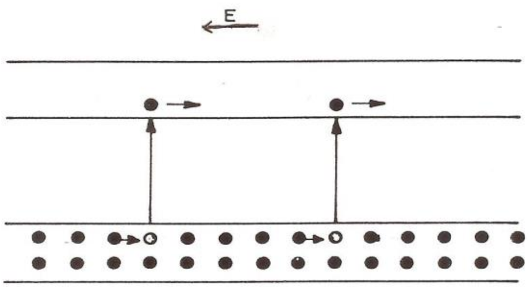


Carrier generation in intrinsic semiconductors:

The number electrons in the conduction band per unit volume of the materials is called the electron concentration, similarly, the number of holes in the valence band per unit volume of the materials is called hole concentration. In general**, the number of charge carriers per unit volume of the materials is called carrier concentration.**

Carrier generation in an **intrinsic semiconductor is due to the breaking of covalent bonds**. Number of covalent bonds breaking is strong function of temperature. Breaking of covalent bond results in the release of an electron into the conduction band. Consequently, an electron energy state becomes vacant in the valence band. This vacant site provides an opportunity for the valence electrons to move under the influence of an electric field. The movement of electron in the valence band will thus equivalent to the movement of vacant site in the opposite direction (Fig.). the vacant site thus created by the transfer of an electron to the conduction band are called 'holes.

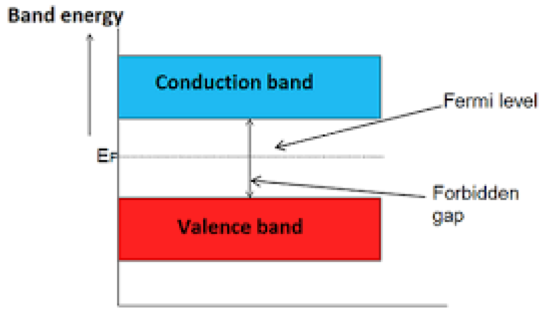
The movement of electrons in the valence band is conveniently explained as equivalent to the movement of holes. Since the direction of movement of these holes is opposite to that of electrons, holes behave as if they are positively charged. Thus, breaking of each covalent band results in the formation of an electron – hole pair, and they contribute to the electrical conductivity of the material. Since the electron and the hole are created together in pair, the number of electrons in the conduction band will be equal to the number of holes in the valence band at any given temperature.



Breaking of covalent bond and generation of electron – hole pairs

Fermi level in an intrinsic semiconductor:

At absolute zero temperature, a semiconductor behaves like an insulator since all the covalent bonds are intact. The valence band is full and the conduction band is empty. At higher temperature, the breaking of covalent bond generates the electron – hole pairs. The number of electrons in the conduction band will be equal to the number of holes in the valence band at any given temperature. It can be shown that for an intrinsic semiconductor, the Fermi level lies at the centre of the band gap and remains invariant with temperature.



We notice that the Fermi factor has a non-vanishing value even for the energy values corresponding to the forbidden energy gap. However, no carriers are present in the forbidden gap since there are no sites available for occupation by the charge carriers.

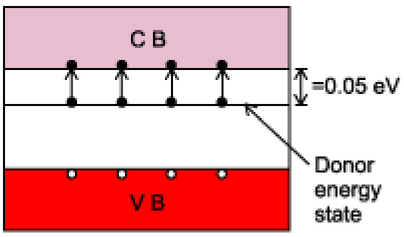
Carrier generation in extrinsic semiconductors:

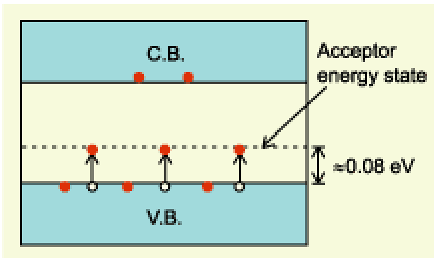
In case of extrinsic semiconductors, the carriers are generated due to two processes, namely

(i) ionization of impurity levels

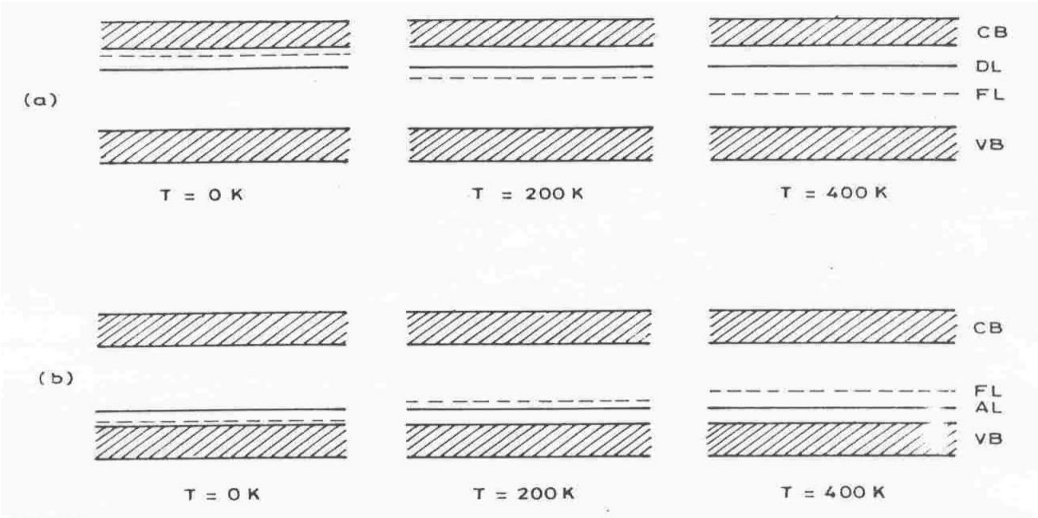
(ii) breaking of covalent bonds

In an n-type semiconductor, for example, the donor levels are close to the bottom of conduction band and require only a small thermal energy to release an electron into the conduction band. This results in a mobile electron in the conduction band and an immobile positive ion of the donor impurity atom. Since the donor levels are very close to the conduction band, a large number of these levels will be ionised even at low temperature. At the same time, there are a few covalent bonds also being broken resulting in the generation of electron - hole pairs. Thus, there will be a large number of electrons (called majority carriers) and a few holes (called minority carriers) contributing to the conductivity of n-type semiconductor. Similarly, the conductivity of a p-type semiconductor will be dominated by majority carrier holes.





Effect of temperature on the Fermi level in an extrinsic semiconductor:



The effect of temperature is illustrated by the energy band diagrams shown in Figure. **In an n-type** semiconductor (fig a), at 0 K all the donor levels will be neutral, i.e., no electrons are contributed to the conduction band by the impurity atoms. The Fermi level, which represents the energy level for which the probability of occupation is half, lies mid-way between the impurity level and the bottom of the conduction band. As the temperature increases, the donor levels will be ionised to a greater extent as compared to breaking of covalent bonds. At elevated temperatures, when most of the donor levels are ionised, the carrier generation due to breaking of covalent bonds will dominate the variation of conductivity with temperature. The Fermi level gradually shifts closer to the centre of the forbidden energy gap and the semiconductor behaves more like an intrinsic semiconductor. This effect is called "**intrinsic effect**". The temperature at which the intrinsic conductivity becomes dominant is called the **transition temperature** at which the extrinsic semiconductor starts behaving like an intrinsic semiconductor.
This transition temperature is also called **maximum device temperature.** It decides the maximum temperature at which the extrinsic semiconductor or the devices made out of the extrinsic semiconductor can be used. For germanium, this temperature is around 70 ^{0}C and for silicon it is about 150^{0}C.

Similarly in a **p-type semiconductor** (fig b), at 0 K all the acceptor levels will be neutral, i.e., no holes are contributed to the valence band by the impurity atoms. The Fermi level, which represents the energy level for which the probability of occupation is half, lies mid-way between the impurity level and the top of the valence band. As the temperature increases, the acceptor levels will be ionised to a greater extent as compared to breaking of covalent bonds. At elevated temperatures, when most of the acceptor levels are ionised, the carrier generation due to breaking of covalent bonds will dominate the variation of conductivity with temperature. The Fermi level gradually shifts closer to the centre of the forbidden energy gap and the semiconductor behaves more like an intrinsic semiconductor.

Conductivity in semiconductors:

Conductivity of an intrinsic semiconductor:

When an electric field is applied to an intrinsic semiconductor, there will be movement of electrons in the conduction band and of holes in the valence band. Hence, there will be two currents, an electron current in the conduction band and a hole current in the valence band. As these currents are constituted by oppositely charged carriers and are in opposite directions, the net current will be sum of the two currents.

If v_e represents the drift velocity of electrons due to the applied field E, the current density due to electrons is given by

$$J_e = nev_e \quad \text{--- (1)}$$

where n is the number of conduction electrons available per unit volume and e is the charge associated with these electrons for the case of a metal.

Similarly, the current density due to holes is given by

$$J_h = pev_h \quad \text{---(2)}$$

The total current density is given by

$$J = (J_e + J_h) = nev_e + pev_h \qquad \text{---(3)}$$

The drift velocity is directly proportional to the applied field E.

i.e. $v \propto E$ or $v = \mu E$ --- (4)

where μ is called the **mobility of carriers**. Hence the above equation becomes

$$J = ne\mu_e E + pe\mu_h E \qquad \text{------ (5)}$$

where μ_e and μ_h are the electron and hole mobilities respectively.

For an intrinsic semiconductor, the number of conduction electrons will be equal to the number of holes. i.e. $n = p = n_i$

Hence $J = n_i eE\,(\mu_e+\mu_h)$ ------- (6)

The conductivity, which is defined as the current density per applied field, is given by

$$\boldsymbol{\sigma = J/E = n_i e\,(\mu_e + \mu_h)} \qquad \text{------ (7)}$$

This equation gives the conductivity of an intrinsic semiconductor.

Expression for the conductivity of an extrinsic semiconductor:

The conductivity of an extrinsic semiconductor can be expressed as

$$\boldsymbol{\sigma = ne\mu_e + pe\mu_h} \qquad \text{--- (1)}$$

where n represents the total number of electrons contributing to conductivity generated by both the mechanisms of carrier generation.

At normal temperature, most of the donor levels will be ionised and each donor atom contributes an electron to the conduction band. Further, the number of electrons generated due to breaking of covalent bonds is very small negligible compared to the number of electrons generated due to ionization of donor levels. Hence, the number of electrons contributing to conductivity may be taken as equal to the donor concentration and the conductivity will be given by

$$\sigma = N_d e\mu_e + pe\mu_h \quad \text{--- (2)}$$

where N_d represents the donor concentration in the semiconductor. Since the donor concentration is quite high compared to the number of covalent bonds broken at room temperature, the second term in the conductivity equation may be neglected

$$\boldsymbol{\sigma_e \approx N_d e \mu_e} \quad \text{--- (3)}$$

Similarly, the conductivity of a p-type semiconductor will be dominated by majority carrier holes and may be represented as

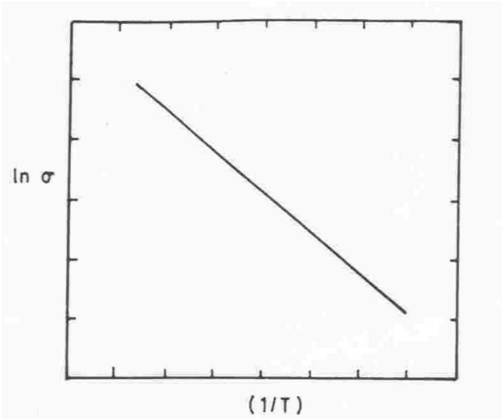
$$\boldsymbol{\sigma_h \approx N_a e \mu_h} \quad \text{---(4)}$$

where N_a is the acceptor concentration in p-type semiconductor.

Effect of temperature on the resistivity of an intrinsic semiconductor:

The conductivity of an intrinsic semiconductor depends strongly on temperature. The higher the temperature, more will be the number of electrons excited to the conduction band. The carrier concentration may be expressed as

$$n = k_1 T^{\frac{3}{2}} exp\left(\frac{-E_g}{2kT}\right) \text{--- (1)}$$



where E_g is the energy gap of the semiconductor and k_1 is a constant. The mobility of electrons and holes generally depend on temperature as

$$\mu = k_2 T^{-3/2} \text{--- (2)} \qquad \text{where } k_2 \text{ is a constant.}$$

Hence the conductivity can be expressed as

$$\sigma = k_1 k_2 e \exp(-Eg/2kT)$$

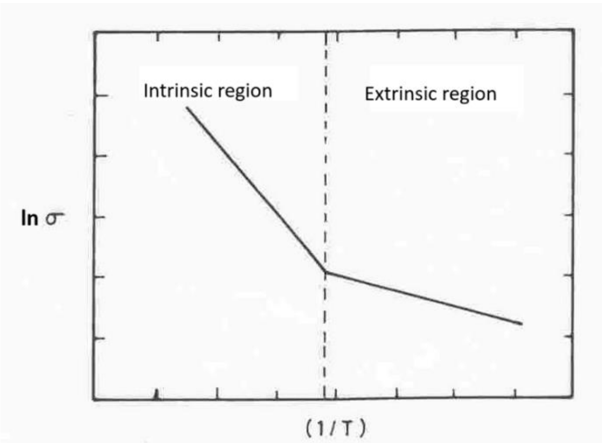
$$\boldsymbol{\sigma = K \exp(-Eg/2kT)} \quad \text{--- (3)}$$

where $K = k_1 k_2 e$

A plot of ln σ versus (1/T) will be a straight line with a slope equal to (Eg/2k). Hence, the band gap energy of the semiconductor can be calculated.

Effect of temperature on the conductivity of an extrinsic semiconductor

The two mechanisms responsible for carrier generation in extrinsic semiconductors, the ionisation of impurity levels predominates at low temperatures as the probability of the covalent bonds being broken will be small. However, as the temperature is increased, more and more covalent bonds will be broken thereby increasing the minority carrier concentration. The temperature dependence of conductivity of an extrinsic semiconductor is shown in Figure. The graph shows a low temperature region, called the extrinsic region, where the conductivity is dominated by the ionisation of impurity levels and a high temperature region, called the intrinsic region, where the carrier generation is dominated by the breaking of covalent bonds.



In the low temperature region,

$$\sigma = K \exp(-E_a/kT)$$

where E_a is the activation energy for the extrinsic conduction and in the high temperature region,

$$\boldsymbol{\sigma = K \exp(-E_g/2kT)}$$

where E_g is the intrinsic band gap of the semiconductor.
The slope of the graph can be used to find the impurity ionization energy (low temp. region) and the intrinsic band gap (high temp. region) of the semiconductor.

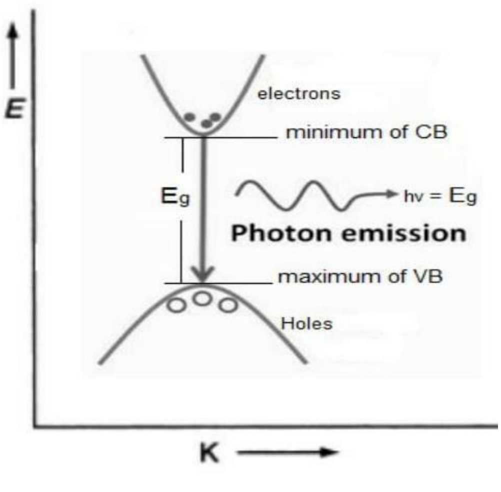
Compare the characteristics of intrinsic and extrinsic semiconductors:

Intrinsic semiconductors	Extrinsic semiconductor
1. It is a chemically pure, structurally perfect and if compound, stoichiometric material	It is a doped semiconductor, obtained by adding a group III or group V element as dopant into an intrinsic semiconductor or nonstoichiometric, if compound.
2. At normal temperature, conductivity is low.	At normal temperature, the conductivity is high.
3. Carrier generation is due to breaking of covalent bonds.	Carrier generation is due ionisation of impurity atoms (forming majority carriers) and breaking of covalent bonds (forming minority carriers).
4. Current carriers, viz., electrons and holes are equal in number	Current carriers are essentially majority carriers at normal temperature
5. Fermi level lies at the centre of the forbidden energy gap at all temperatures	At 0 K, the Fermi level lies close to the conduction band in n-type semiconductor and close to the valence band in p-type semiconductor. Fermi level moves towards centre of the band gap as temperature increases.
6. Hall coefficient is negative due to higher mobility of electrons.	Hall coefficient is negative for n-type semiconductor and positive for p-type semi-conductor.

<u>Direct And Indirect Band Gap Semicondcutors</u>:
Based on the structure of energy bands and type of energy emission, semiconductors are classified into two types.

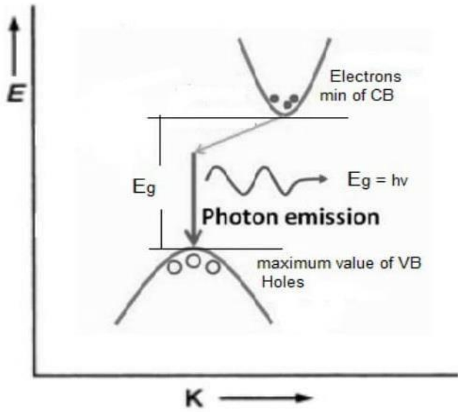
1. Direct band gap semiconductor
2. Indirect band gap semiconductor

Direct Band Gap Semiconductor:
The maximum of VB and the minimum of CB exists at the same value of wave number (k), Such semiconductors are called direct band gap semiconductor. Where, Wave number (k) represents the propagation vector and is related to the momentum of the carrier. In this, electron can directly excite or de-excite by the absorption or emission of photon and there is no phonon involvement in the process of excitation and de-excitation. In this type during the recombination of electrons and holes, the energy



difference Eg is released as a photon of light. This process is known as radiative recombination. These semiconductors are used to fabricate LEDs and laser diodes. These are usually from the compound semiconductors.
Ex: GaAs, InP, …….

Indirect Band Gap Semiconductor:
The maximum value of VB and the minimum value of CB exists at the different values of wave number (k). Such semiconductors are called Indirect Band Gap Semiconductor. If an electron goes from the top of the valence band to the bottom of the conduction band, it has to change its energy as well as wave-vector K (momentum). For conservation of momentum and energy, there is the involvement of phonon in the process. If there is de-excitation of the electron, then not all the energy will be emitted in the form of the photon but some energy is emitted in the form of phonons i.e. some part is transferred to the lattice, and the lattice will vibrate and generate heat. This process is known as non-radiative recombination. So indirect bandgap semiconductor is not suitable for light emission. These are usually from the elemental semiconductors. ***Ex: Si, Ge***.



The band gap of a semiconductor can be deduced from optical absorption studies. Such studies also give us information whether the band gap is a direct one or an indirect one. When radiation of suitable energy is incident on a semiconductor sample of thickness x, the intensity of radiation transmitted is given by $\mathbf{I = I_o \exp(-\alpha x)}$
where Io is the incident intensity and α is the absorption coefficient. In a typical plot of the variation of α with the energy of the incident radiation**,**

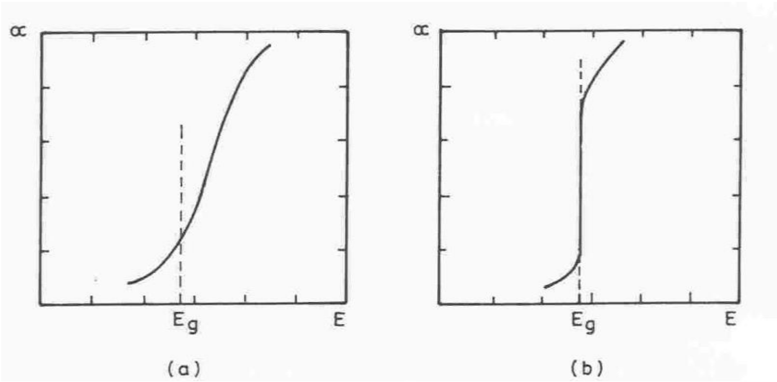


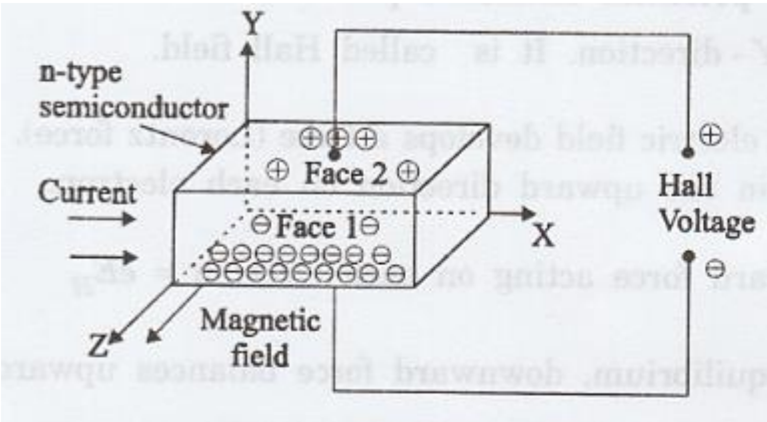
Fig.2. Variation of absorption coefficient as a function of energy for (a) indirect band gap semiconductor and (b) direct band gap semiconductor.

<u>**HALL EFFECT:**</u>
If a current carrying conductor or semiconductor sample is placed in a transverse magnetic field, an electric field is induced in the sample in a direction perpendicular to both the

current and the magnetic field. This phenomenon is called Hall effect. The electric field generated is called Hall field and the corresponding voltage is called Hall voltage.

Consider a rectangular slab of an n-type semiconductor sample of length l, breadth b and thickness t. Let I be the current flowing through the sample along positive x-direction. A magnetic field B is applied perpendicular to the direction of current flow, say along positive z-direction. The effect of the magnetic field is to exert a force on the current carriers, namely electrons, flowing in the negative x-direction. As a result of this force, the electrons will be forced downwards in the negative y-direction. This results in the lower face (Face 1 in the figure) becoming negatively charged with respect to the top face (Face 2). This constitutes the Hall field E_H. The electrons entering later will experience two forces –

(i) force Bev due to magnetic field, acting downwards and
(ii) force eE_H due to Hall field, acting upwards.



In equilibrium, these two forces will be equal and opposite.

Hence $eE_H = Bev$ ---- (1)

where v is the drift velocity of the electrons.

Or $E_H = Bv$ ---- (2)

The current density J can be expressed in terms of concentration of carriers n and their drift velocity as

$J = nev$ or $v = J/ne$

Therefore, we can write the equation (2) as

$\mathbf{E_H = BJ/ne}$ **------ (3)**

Or $\mathbf{E_H = R_H BJ}$ **------ (4)**

Where $\mathbf{R_H} = \frac{\mathbf{1}}{\mathbf{ne}}$ is called **Hall coefficient**, which is a measure of strength of the Hall field induced.

For a given semiconductor, the **Hall coefficient R_H** is defined as the **Hall field generated in the material per unit magnetic field applied when the current density is unity**.

i.e, $\mathbf{R_H} = \frac{\mathbf{E_H}}{\mathbf{B\,J}}$ ---- (5)

In case of n-type semiconductor,

$R_H = -\frac{1}{ne}$ ---- (6)

since the Hall field developed is in negative Y- direction.

For a p-type semiconductor, R_H will be positive and is given by

$$R_H = \frac{1}{Pe} \qquad \text{---- (7)}$$

Expression for Hall voltage and carrier concentration: Let the rectangular slab has length l, breadth b and thickness t and a current I flowing through it results in a current density J across the cross-section A= bt. Let B be the magnetic field acting along the thickness t.

Then, $E_H = V_H / b$

Cross sectional area, A=b x t,

Therefore, current density

$$J = \frac{I}{A} = \frac{I}{b\,t}$$

Therefore equation (3) can be written as

$$\frac{V_H}{b} = \frac{B}{ne}\frac{I}{bt}$$

where V_H is the Hall voltage generated across the breadth and I is the current flowing along the length.

Or $$\boldsymbol{n = \frac{BI}{etV_H}} \qquad \textit{and} \qquad \boldsymbol{V_H = \frac{BI}{net}}$$

These equations are valid for the case of free electrons moving with steady drift velocity. However, in practice, assuming average drift velocity for carriers, a correction factor of $\frac{3\pi}{8}$ is included. Hence

$$\boldsymbol{n = \frac{3\pi}{8} \cdot \frac{BI}{etV_H}} \qquad \text{and} \qquad \boldsymbol{V_H = \frac{3\pi}{8} \cdot \frac{BI}{net}}$$

Here, n represents the number of charge carriers per unit volume of the semiconductor.

Expression for Mobility:

For an n-type material, the conductivity is given by $\sigma_n = ne\mu_e$

$$\therefore\ \mu_e = \frac{\sigma_n}{ne} = \boldsymbol{\sigma_n R_H}$$

Similarly, for a p -type material, $\mu_h = \frac{\sigma_h}{pe} = \boldsymbol{\sigma_p R_H}$

Application of Hall effect:

1. Determination of semiconductor type
2. Calculation of carrier concentration
3. Determination of mobility
4. Measurements of magnetic flux density
5. Hall effect multiplier

<u>The p-n Junctions:</u>

A p-n junction, is a system of two semiconductors in physical contact, one with excess of electron (n-type) and the other with excess of holes (p-type).

The p-n junctions are among most simple semiconductor devices. They are use full devices in the electronic circuits to perform functions like rectification, amplifications, switching, etc. A p- n junction may be obtained by different methods and the junctions are known by the of methods of fabrication. They are:

a) Grown – in – junction
b) Alloy junction
c) Diffused junction
d) Ion implanted junction
e) Deposited junction

Unbiased p-n junction.

Consider a junction between an n-type and p-type semiconductor which has just been formed. The n-type semiconductor contains some of the semiconductor atoms replaced by donor atoms of group V. Similarly, the p-type semiconductor contains some of the semiconductor atoms replaced by acceptor atoms belonging to group III. The concentration of the donor atoms may be different from the concentration of acceptor atoms in a given junction.

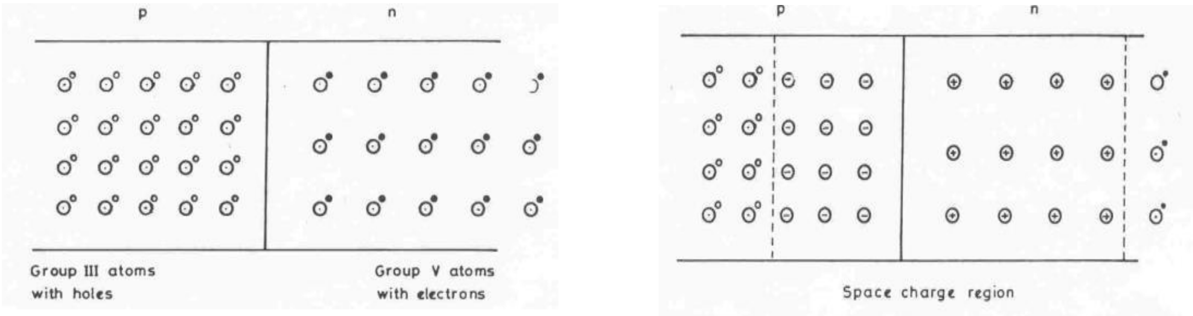


Fig. Formation of a p-n junction (a) before charge transfer and (b) after charge transfer.

Fig. shows a p-n junction so formed where only the impurity atoms are shown. A n-type impurity atom is shown to have an additional electron weakly attached to it. Similarly, p-type impurity atoms are associated with holes. When the junction is formed, the electrons from the n-type region diffuse into the p-type region because of the large concentration of electrons on the n-side and the lack of it on the p-side. Hence, the electrons from the donor atoms close to the junction cross the junction to combine with the holes on the p-side. When the electrons leave the donor atoms, they leave behind positively charged donor ions. Similarly, the acceptor atoms become negatively charged acceptor ions when the electrons recombine with the holes. Consequently, there is positively charged region of immobile donor ions close to the junction on the n-side and a negatively charged region of immobile acceptor ions on the p-side. These are referred to as **space charge regions**. They are also called **charge depletion regions** since there are no mobile charges in these regions. However, such diffusion of carriers stops when the space charge produced is sufficient to prevent the carriers from crossing over. The potential existing due to the space charge regions at equilibrium is called the **barrier potential**. Since the total charge density in the space charge region on either side of the junction is equal, we have

$$N_d X_n = N_a X_p$$

where N_d and N_a represent the concentration of impurities in the n-type and p-type materials, X_n and X_p are the widths of the charge depletion regions on the two sides. This is called the **charge**

neutrality condition. Thus, we see that if the concentration of impurities is higher on any side, the width of the depletion region on that side will be proportionately smaller.

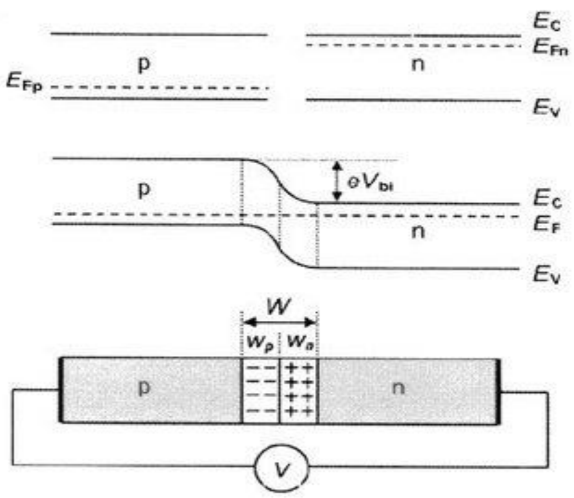
Energy level diagrams and the formation of a potential barrier in a p-n junction

Figure below shows the energy level diagrams for the p-n junction before and after the diffusion of carriers. When the equilibrium is established after the transfer of carriers, the Fermi levels on the n-side and p-side will be levelled since the Fermi level represents the energy level where the probability of occupation is half and is same on either side of the junction close to it. As a result, the energy bands on the p-type side will be at a higher energy level as compared to those on the n-side.

An electron at the bottom of the conduction band on the n-side has to acquire an energy equal to eV_B to cross over to the p-side, where V_B represents the barrier potential. It can be shown that the barrier potential is related to the concentration of impurities on the two sides of the junction and is given by

$$V_B = KT \log_e \frac{N_a N_d}{n_i^2}$$

The width of the charge depletion regions is also a function of the impurity concentration and decreases with increase in impurity concentration.



Energy level diagrams for a p-n junction (a) before the formation of the junction and (b) after the formation of the junction.

Forward biasing a p-n junction.

If the p-type material of the junction is made positive with respect to the n-type, the junction is said to be **forward biased.**

Consider a p-n junction with a forward bias V applied to it. Since the n-type region is made negative, the applied field assists the majority carrier electrons to cross over to the p-side of the junction. Energetically, this can be represented by

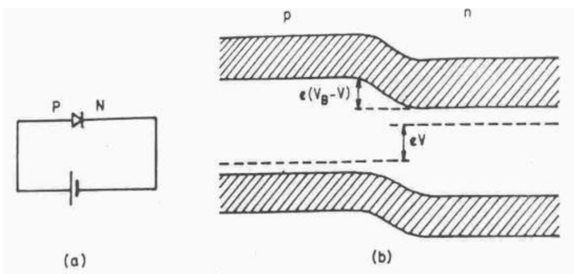


Fig. The energy level diagram for forward biased p-n junction an energy band diagram of the p-n junction with all the energy levels (including the Fermi level) raised by an energy eV where V is the applied forward bias. In other words, the effective barrier is now reduced to a value (VB-V) which results in an increased diffusion current of majority carriers. However, the minority carrier current due to the drift remains unaltered. Thus, a net current of majority carrier electrons will flow from n-type region to the p-type region. Also flowing will be a net current of majority carrier holes from p-type to n-type region which, with the electron current, constitutes the total current across the junction. As a result, the p-n junction will offer a low resistance to the flow of current when forward biased.

Reverse biasing a p-n junction

The junction is said to be **reverse biased** when the p-type region is made negative with respect to the n-type region. The applied voltage now adds to the barrier potential to present an increased barrier (V_B + V) to the diffusion of majority carriers. If the reverse bias is large enough, the majority carriers will be prevented from moving across the junction. The minority carrier current of electrons from p-type to n-type and of holes from n-type to p-type will now be constituting the total current across the junction. Since the drift current is independent of the barrier height, this reverse current is called the reverse saturation current. Thus, the p-n junction offers high resistance to the flow of current when reverse biased. The voltage–current characteristics of a p-n junction under forward and reverse bias conditions is shown in Fig. The total current across the junction is given by the diode equation,

$I = I_0 [\exp(eV/kT) - 1]$, where V is the applied bias.

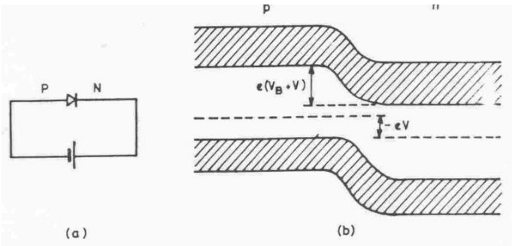


Fig. The energy level diagram for reverse biased p-n junction.

Breakdown in a p-n junction:

In a reverse biased p-n junction, as the applied voltage is increased the reverse current remains practically constant. As the voltage is increased further, at a particular voltage, the current begins to increase very rapidly. The reverse biased p-n junction which was offering a high resistance to the flow of current suddenly starts conducting heavily. This phenomenon in which the current increases very rapidly at a particular reverse voltage is called breakdown. The mechanism of breakdown is due to impact ionisation of lattice atoms in the depletion region by the energetic minority carriers. For example, an electron from the p-side may acquire enough kinetic energy from the applied electric field to cause ionising collision with lattice atoms. This results in the breaking of covalent bonds leading to the generation of electron-hole pairs. The carriers so generated are further accelerated by the applied field and may cause ionisation events thereby resulting in an avalanche process and hence a large current. This type of breakdown is called “**Avalanche breakdown**” and is normally observed in lightly doped p-n junctions.

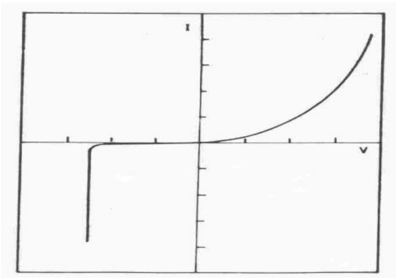


Fig. Current–voltage characteristics for a p-n

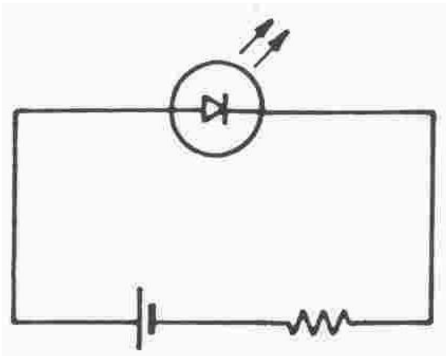
Semiconductor Devices:

1. Light Emitting Diode (LED):

A light-emitting diode is a semiconductor diode that emits light when forward biased. LEDs are made up of direct band gap semiconductors like GaAs, GaP, GaAsP etc.

Basic Principle: When a p-n junction is forward biased, the majority carriers (electrons) diffuse from the conduction band of n-type region into the conduction band of the p-type region. This is referred to as **minority carrier injection** since electrons are minority carriers in the p-type region. This disturbs the equilibrium between electrons and holes in the p-type region and in order to regain equilibrium concentration, the electrons recombine with holes thereby releasing energy. In LEDs made up of direct band gap semiconductors a large fraction of the energy is released as light. The process of emission of light from a forward biased p-n junction is called **'injection electroluminescence'.**

When electrons in the conduction band recombine with holes in valence band, a spontaneous emission of the radiation takes place. The energy of the emitted photon(hv) is equal to the energy gap (E_g) of materials, that is



$$E_g = h\upsilon \quad \text{---------(1)}$$

where h is the planks constant υ is the frequency of the emitted radiation.

Substituting the value of $\upsilon = \frac{c}{\lambda}$ in equation (1) we get

$$E_g = \frac{hc}{\lambda} \quad \text{---------(2)}$$

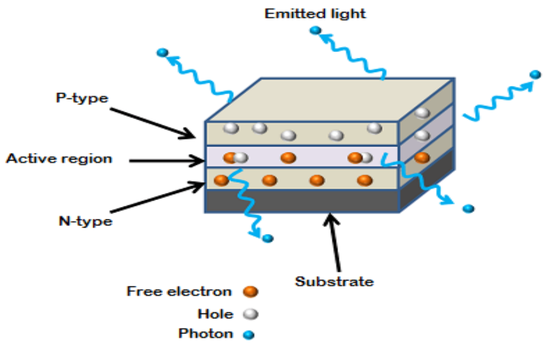
where c is the velocity of light and λ is the wavelength of light. Equation (2) can also be written as

$$\lambda = \frac{hc}{Eg} \quad \text{----------(3)}$$

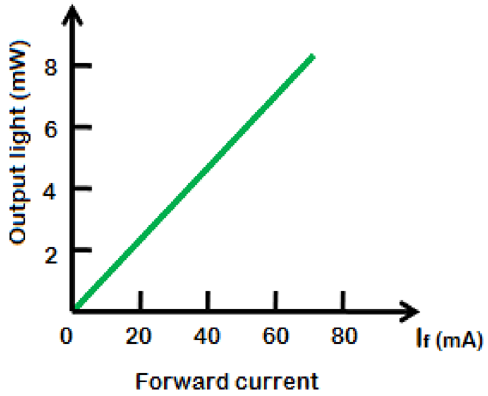
This indicates that wavelength of emitted photons depends on the energy gap in the semiconductor. Thus, the energy gap of a semiconductor plays a major role in selecting a suitable material for LED applications.

LED construction: LED is made of three layers i.e. P-type semiconductor layer, N-type semiconductor layer and active region. The active region is also known as the depletion region. The electrons and holes recombine in this region.

Most LEDs are generally based on gallium semiconductor and are constructed using Gallium Arsenide (GaAs), Gallium Arsenide Phosphide (GaAsP), Gallium Phosphide. A commercial light emitting diode is made out of a suitable material and the diode is embedded in a clear plastic cover with a parabolic reflector provided at the back to direct the emitted radiation.



Construction of LED



LED is a forward bias p-n junction as shown in fig which emits visible light when suitably biased for the duration of the forward biasing, the charge carriers, that is electrons and holes are injected respectively into the anode and cathode region. The recombination of the charge carriers that is, the electron from the n-side and the holes from the p-side, takes place at the junction. Throughout the recombination the difference in the energy is given up in the form of heat and light radiation that is photons. Thus, the diode current controls **the electroluminuous efficiency of the LED.**

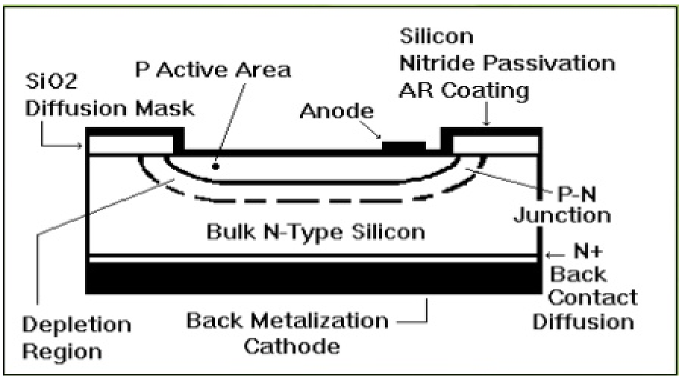
Output characteristics of LED: The amount of output light emitted by the LED is directly proportional to the amount of forward current flowing through the LED. More the forward current, the greater is the emitted output light. The graph of forward current vs output light is shown in the figure.

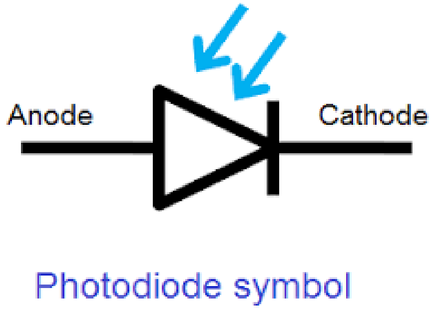
Applications of LED: The various applications of LEDs are as follows:
Light emitting diodes are used in visual display units, calculators, clocks, panel meters, pilot lamps, etc.
They are used in optical communication applications using fiber optics like data transmission and telephones.
They are also used in card and paper tape readers, burglar alarms, etc.

2. <u>Photodiode:</u> A photodiode is a p-n junction semiconductor device that converts light energy into electrical energy, when illuminated by light in reverse bias condition. It is also referred as photo-detector, photo-sensor, or light detector.

Principle: When a p-n junction is reverse biased and is illuminated, majority and minority carriers are optically generated. The majority carriers created do not increase the current, since the junction is reverse biased. But the minority carriers so created are swept across the junction (depletion region) by drift, provided they are generated within a diffusion length of the depletion region. The creation of minority carriers by using light is called **optical injection** and these generated carriers are called **excess carriers**, because they are excess to those generated thermally. Under reverse bias the amount of diffusion remains negligible but minority drift current increase dramatically with increase in light intensity.

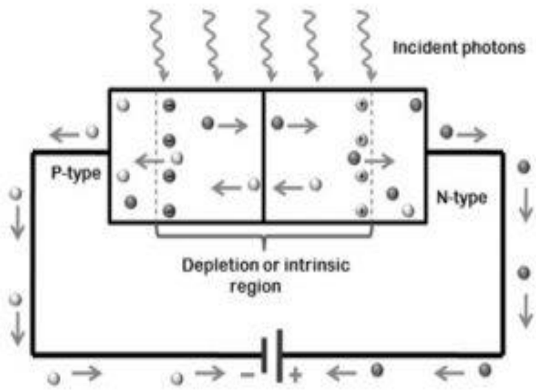
Construction of Photo diode: Photodiode is a PN junction diode. Its construction is shown is figure.

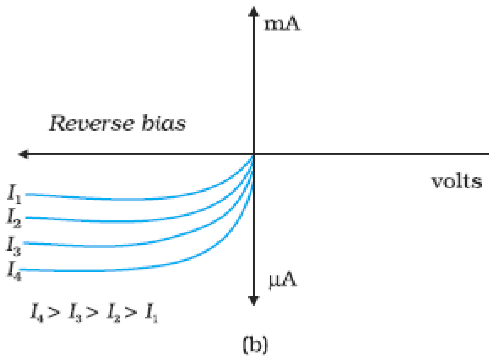




The photo diode consists of a p-n junction separated by a wide depletion region. Small metal contacts are applied to the front surface and the entire back is provided with metal contact. The junction needs to be very close to the surface in the photodetector. The distance of the junction from the surface must be less than the diffusion length. So, the p-region is made thin. Upper p-layer surface is coated with an antireflecting coating to reduce the reflection of light. Silicon dioxide (SiO_2) patches are used to insulate the surface of one device from its neighbour. The diode is embedded in clear plastic with all sides painted black except a small aperture through which light is concentrated on to the junction.

Working of Photodiode: When the photodiode is reverse biased at less than its breakdown voltage and is in not exposed to light (dark condition), a very small reverse current flows through the device that is termed as dark current. It is called so because this current is totally the result of the flow of minority carriers and flows when the device is not exposed to radiation.





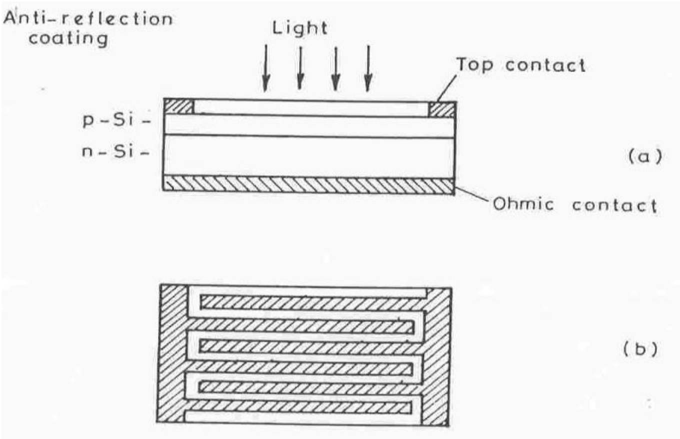
When the active region (junction) of the device is illuminated with light, electron–hole pairs are formed. These electron and holes get separated under the applied bias and drift across the junction. This results in a reverse current. As the intensity of the incident light increases more and more electron- holes are produced, which increases the reverse current as shown in the I—V curve. So, the intensity of light energy is directly proportional to the current through the device. The first curve represents the dark current that generates due to minority carriers in the absence of light. Thus, the photodiode is useful for detecting as well as measuring the intensity of incident radiation.

The diode detects over a desired wavelength range with enough sensitivity to enable practical use.

Applications: To detect and measure visible, IR UV radiation. To convert optical signal into electrical signal, switching application, light operated switches, burglar alarms, reading optical tracks of recorded sound etc.

2. Solar Cell: Solar cell is a p-n junction that converts the light energy (solar energy) into electrical energy. This device works on the principle of photovoltaic effect. (The effect in which light energy is converted to electrical energy in certain semiconductor materials is known as photovoltaic effect).

Construction: The construction of a solar cell is shown in the figure. The solar cell consists of a p-n junction of appropriate thickness of p and n regions. The p- region is made thin enough to allow sufficient number of photons to reach the junction. An antireflection coating is given on top of the p- region to reduce possible reflection of the incident radiation. The top metal contact is made in the form of a grid to ensure effective collection of carriers. The metal contact completely covers the lower face.



Construction of a solar cell: (a) cross-sectional view b) top view.

Working of a solar cell: When a solar cell is exposed to sunlight, the photons provide energy to the p-n junction, creating electron-hole pairs. This light disrupts the thermal equilibrium condition of the junction, encouraging the free electrons to move to the n-side of the junction. The holes move to the junction's p-side in a similar pattern. As a result, free electrons on the n-type side fail to move past the junction due to a potential barrier. The same barrier potential of the junction blocks the newly created holes, causing an increased concentration of electrons on one side (at the n-type junction) and holes on the other side. This process allows the p-n junction to operate as a battery cell.

Under open circuit condition (i.e., when no load is connected across the p-n junction), the net current is zero and hence, the photovoltaic potential or the **open circuit photovoltage, V_{oc}**, is observed across the p-n junction. The current through the solar cell when no load (R=0) is connected is called the **short circuit current I_{sc}**.

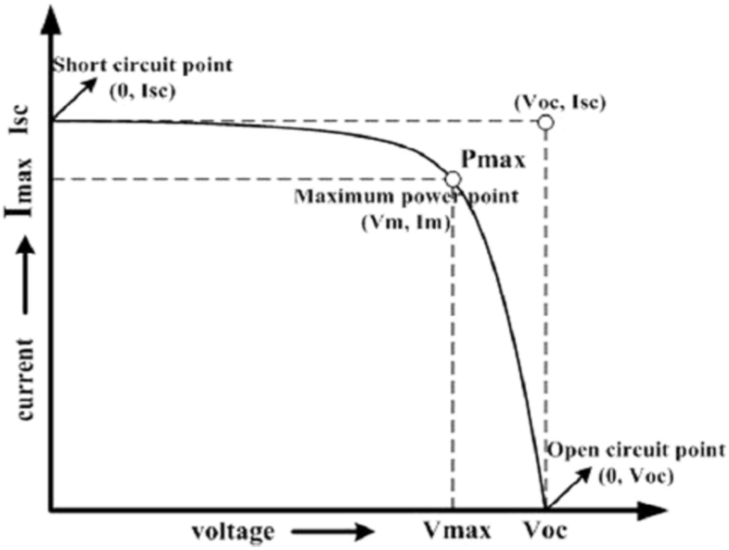
When a load resistance R_L is connected across the illuminated p-n junction, the current and voltage values across the load R_L are measured. The power that can be drawn from the p-n junction can be calculated as P = VI.

when $R_L = 0$, $V = 0$, $\therefore P = 0$

when $R_L = \infty$, $I = 0$, $\therefore P = 0$

We find that the output power is maximum for a certain optimum load resistance.

$P_{mpp} = V_m.I_m$ where Vm and Im are the voltage and current values corresponding to maximum power point (mpp). When the diode is used for converting radiant energy into electrical energy, the diode should be operated with this optimum load resistance. Since power can be delivered to an external circuit by an illuminated junction, it is possible to convert solar energy into electrical energy.



Efficiency and fill factor of a solar cell :

The **conversion efficiency** of a solar cell is defined as

$$\eta = \frac{\text{Output electrical power}}{\text{Input radiant power}}$$

The efficiency of a solar cell is also expressed in terms of **quantum efficiency** which is defined as

$$\eta * = \frac{\text{number of electron} - \text{hole pairs generated}}{\text{number of photons absorbed}}$$

A well-made silicon solar cell can have a conversion efficiency of 15% to 20% (η = 0.15 to 0.2).

Fill factor is a measure of the maximum power that can be drawn from a solar cell and is given by

$$\text{Fill Factor, FF} = \frac{(\text{Vm}).(\text{Im})}{(\text{Voc})\text{x}(\text{Isc})}$$

PROBLEMS:

1. Calculate the conductivity of silicon doped with 10^{21} atoms m^{-3} of boron if the mobility of holes is 0.048 $m^2v^{-1}s^{-1}$.

   Solution: Conductivity of the extrinsic semiconductor with an acceptor concentration of Na is given by

   $\sigma = N_a e \mu_h = 10^{21} \times 1.6 \times 10^{-19} \times 0.048 = \mathbf{7.68\ ohm^{-1}m^{-1}}$

2. Calculate the resistivity of intrinsic germanium if the intrinsic carrier density is 2.5 x 10^{19} m^{-3} assuming electron and hole mobilities of 0.38 and 0.18 $m^2v^{-1}s^{-1}$respectively.

   Solution: Resistivity is given by

   $\rho = 1/\sigma = 1 / [ne(\mu e + \mu h)]$

   $= 1/ [2.5\times 10^{19} \times 1.6\times10^{-19} \times(0.38+0.18)] = 0.45$ ohm m

3. Calculate the electron and hole densities in n-Si doped with 10^{19} donors m^{-3} if the intrinsic carrier concentration in silicon is $1.4 \times 10^{16} m^{-3}$.

Solution:

Density of electrons, $n = 10^{19} + 1.4 \times 10^{16} = 10^{19}\ m^{-3}$

Density of holes, $p = n_i^2/ n$ where n_i is the intrinsic concentration.

$p = (1.4 \times 10^{16})^2 / 10^{19}$

$= 1.96 \times 10^{13}$.

4. Calculate the conductivity of intrinsic germanium at 300 K. Given that intrinsic carrier concentration at 300 K is $2.4 \times 10^{19}\ m^{-3}$ and the mobilities of electrons and holes are $0.39\ m^2v^{-1}s^{-1}$ and $0.19\ m^2v^{-1}s^{-1}$ respectively. What is the conductivity after doping germanium with donor – type impurity to the extent of one atom per 10^8 germanium atoms? Given that the number of Ge atoms/m^3 are 4.4×10^{28}

.Solution:

Conductivity of intrinsic germanium is

$\sigma_I = n_i e (\mu_e + \mu_h) = 2.4 \times 10^{19} \times 1.6 \times 10^{-19} \times (0.39 + 0.19) = 2.23\ ohm^{-1}\ m^{-1}$

Conductivity of n-type germaniumis

$\sigma_n = ne\mu_e + pe\mu_h$

$n \approx N_d \approx 4.4 \times 10^{20} m^{-3}$; $p = n_i^2/ N_d = 1.3 \times 10^{18} m^{-3}$

Since p is very small compared to n, the second term in the conductivity equation may be neglected.

$\therefore \quad \sigma_n = N_d e \mu_e = 27.46\ ohm^{-1} m^{-1}$

5. The electrical conductivity of an intrinsic semiconductor increases from 19.96 $ohm^{-1}m^{-1}$ to 79.44 $ohm^{-1}m^{-1}$ when the temperature is increased from 60 to 100^{o}C. Find the band gap energy of the semiconductor.

Solution: The electrical conductivity of an intrinsic semiconductor is given by

$\sigma = k.\ \exp(E_g /2kT)$

Neglecting its weak temperature dependence, k can be considered to be a constant. A graph of ln σ versus (1/T) will have a slope given by

$[\Delta(\ln\sigma)] / [\Delta(1/T)] = -E_g/ 2k$

$$E_g = -\frac{(\ln \sigma_2 - \ln\sigma_1)}{(1/T_2 - 1/T_1)} \times 2k$$

$= 1.184 \times 10^{-19}\ J = 0.74\ eV.$

6. A semiconductor sample of thickness 1.2×10^{-4}m is placed in a magnetic field of 0.2T acting perpendicular to its thickness. Find the Hall voltage generated when a current of 100 mA passes through it. Assume the carrier concentration to be $10^{23} m^{-3}$.

Solution:

Hall voltage generated is given by the formula $V_H = \underline{3\pi BI/}8net$

$$V_H = \frac{3 \times 3.14 \times 0.2 \times 100 \times 10^{-3}}{8 \times 10^{23} \times 1.6 \times 10^{-19} \times 1.2 \times 10^{-4}}$$

$$V_H = 0.0123 \text{ Volt or } 12.3 \text{ mV}$$

7. Intrinsic silicon has a carrier concentration of $1.1\times10^{16} m^{-3}$. If the mobilities of electrons and holes are 0.17 and 0.035 $m^2V^{-1}s^{-1}$ respectively at room temperature, compute the resistivity of silicon.

Solution: Conductivity is given by

$$\sigma_i = n_i e (\mu_e + \mu_h)$$

$$= 1.1\times10^{16} \times 1.6\times10^{-19} (0.17 + 0.035)$$

$$\sigma_i = 3.608 \times 10^{-4} \text{ ohm}^{-1} \text{ m}^{-1}$$

Resistivity $\rho = 1/\sigma = 2.77 \times 10^3$ ohm m

8. The compound gallium arsenide has an intrinsic conductivity of 10^{-6} ohm^{-1} m^{-1} at 20^o C. How many electrons have jumped the forbidden energy gap? [Given: $\mu_e = 0.88$ $m^2V^{-1}s^{-1}$ and $\mu_h = 0.04$ $m^2V^{-1}s^{-1}$]

Solution:

Number of electrons that have crossed the forbidden energy gap is the number of electrons contributing to conductivity, which is the carrier concentration, n_i.

$$n_i = \sigma/e(\mu_e + \mu_h)$$

$$= 1\times10^{-6}/1.6\times10^{-19}(0.88 + 0.04)$$

$$= 6.79\times10^{12} m^3.$$

9. An electric field of 100 V m^{-1} is applied to a sample of n- type semiconductor having a Hall coefficient – 0.0125 m^3C^{-1}. Determine the current density in the sample if the electron mobility is $0.36m^2V^{-1}s^{-1}$.

Solution: Current density $J = nev = ne\mu E = -\mu E/R_H$

$$= 0.36 \times 100/0.0125$$

$$= 2880 \text{ Am}^{-2}.$$

10. The resistivity of germanium at 20^o C is 0.5 ohm m. What will be its resistivity at 40^oC if the band gap of germanium is 0.7 eV?

Solution: We have the relation,

$$[\Delta(\ln\rho)]/[\Delta(1/T)] = E_g / 2k$$

$$\Delta(\ln\rho) = E_g.\Delta(1/T)/2k$$

$$(\ln \rho_{40} - \ln \rho_{20} = -0.885$$

$$\ln \rho_{40} = \ln \rho_{20} - 0.885 = -0.693 - 0.885 = -1.578$$

$$\rho_{40} = 0.206 \text{ ohm m.}$$

11. Sample of silicon is doped with 10^{22} phosphorus atoms m^{-3}. What would you expect to measure for its resistivity? What Hall voltage would you expect in a sample 100 μm thick when a current of 1 mA is passed perpendicular to a magnetic field of 0.1T

[Given: $\mu_e = 0.07 m^2V^{-1}s^{-1}$].

Solution:

Conductivity $\sigma = N_d e \mu_e = 10^{22} \times 1.6\times10^{-19}\times0.07 = 112\ ohm^{-1}m^{-1}$.

Resistivity $\rho = 1/\sigma = 8.93 \times 10^{-3}$ ohm m (**Ans**).

Hall voltage $V_H = 3\pi BI/8net$

$=(3\times3.14\times0.1\times1\times10^{-3})/(8\times10^{22}\times1.6\times10^{-19}\times100\times10^{-6})$

$= 0.735$ mV (**Ans**).

Problems

1. Calculate the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K.
2. Find the temperature at which there is 1% probability that a state with an energy 0.5 eV above Fermi energy is occupied.
3. A sample of silicon semiconductor is doped with 10^{22} phosphorous atoms. Calculate its conductivity if mobility of electrons is 0.07 m^2/V/s. What is the Hall voltage if this semiconductor with a thickness of 100μm and carrying a current of 1 mA is placed perpendicular to a magnetic field of 0.1T.?

Question Bank: Semiconductors

1. What is a semiconductor? Distinguish intrinsic and extrinsic semiconductor.
2. What are direct and indirect band-gap semiconductors? Explain.
3. What is an intrinsic semiconductor? Explain the mechanism of carrier generation in intrinsic semiconductors.
4. What are intrinsic semiconductors? Obtain an expression for the conductivity of an intrinsic semiconductor.
5. Explain the effect of temperature on the conductivity of an intrinsic semiconductor.
6. What are extrinsic semiconductors? Describe the mechanisms of carrier generation in extrinsic semiconductors.
7. Obtain an expression for the conductivity of an extrinsic semiconductor.
8. Explain the effect of temperature on the conductivity of an extrinsic semiconductor.
9. Explain the effect of temperature on the Fermi level in an intrinsic and extrinsic semiconductor.
10. Write any four differences between conductors and semiconductors. Increase in temperature decreases the resistivity of a semiconductor, why?
11. State Hall Effect? Obtain an expression for the Hall voltage and Hall coefficient of an n-type semiconductor.
12. Describe the experimental method of determining the carrier concentration in an extrinsic semiconductor using Hall effect.
13. Illustrate the formation of the depletion region in a PN junction, how it varies in forward and reverse biased conditions and study its V-I characteristics under biased conditions.
14. Explain the principle, construction and working of light emitting diode.
15. Explain the principle, construction and working of photo diode.
16. Write a note on solar cell.
17. Explain the principle, construction and working of a solar cell.